texopt Scoring Model v2 (2026-10-14)

Hard-constraint term L (model 1) and aesthetic term A (model 2)

Notation. x_0 = original manuscript; x' = candidate; $a = (a_1, \ldots, a_n)$ = action sequence; $R(x')$ = perception of x' *after a full re-compilation* (page count, log defects, rendered page images); r = requirement spec; $b = r.\text{page_limit}$ ($b = \varnothing$ means "no page limit was requested"). Every quantity is a function of $R(x')$, i.e. compile-then-score globally; no per-fragment scoring.

1. Model 1: feasibility L + intervention cost I

1.1 Hard-constraint set $\mathcal{H}$ (all indicators are 0/1)

$$\ell_{\text{compile}}(x') = 1 - \mathbf{1}[R(x') \text{ compiles } \wedge \text{ PDF produced}]$$

$$\ell_{\text{page}}(x') = \begin{cases} \mathbf{1}[N_{\text{pages}}(x') > b]\,, & b \neq \varnothing \\ \text{the variable does not exist}, & b = \varnothing \end{cases} \qquad \text{(directive 2: 0/1, no middle ground)}$$

$$\ell_{\text{spec}}(x') = \frac{1}{K}\sum_{j=1}^{K} \mathbf{1}\big[p_j(x') \neq p_j^{\text{req}}\big] \quad \text{(font size / margins / fleqn / TOC / header — only the items named by } r\text{)}$$

$$\ell_{\text{fig}}(x') = \mathbf{1}[\text{fingerprints}(x') \neq \text{fingerprints}(x_0)]$$

1.2 Definition of L (revised)

$$\boxed{L(x') = \sum_{i \in \mathcal{H}} w_i\, \ell_i(x') = w_c\, \ell_{\text{compile}} + w_p\, \ell_{\text{page}} + w_s\, \ell_{\text{spec}} + w_f\, \ell_{\text{fig}}}$$

Each $\ell_i \in \{0, 1\}$: being 0.2 pages over the limit and being 5 pages over score identically. If $b = \varnothing$, ℓ_{page} is *absent from the sum* (not zero — undefined), implemented as a single guard `if req.page_limit:`. $L = 0$ is the minimum.

1.3 Content preservation: removed from $\mathcal{H}$ (directive 1)

It has become a *soft* term instead of a hard constraint, because the later *semantic-polishing* stage is allowed to rewrite body text:

$$\widetilde{B}(x) = \text{normalized body (layout commands stripped, whitespace collapsed)}$$

$$\text{Keep}(x') = \frac{\left|\text{multiset}\left(\widetilde{B}(x')\right) \cap \text{multiset}\left(\widetilde{B}(x_0)\right)\right|}{\left|\widetilde{B}(x_0)\right|}, \qquad D_{\text{content}}(x') = 1 - \text{Keep}(x')$$

Weight $\beta = \texttt{CONTENT_PENALTY} = 50$: a strong preference, not a prohibition. The semantic stage adds the hard floor $\text{Keep}(x') \geq \kappa$ later.

Honest boundary: D_{content} is a *magnitude-of-change proxy*, not a proof of semantic equivalence (multiset overlap flags paraphrase as change — intentionally conservative).

1.4 Intervention cost I

$$I(x') = 0.5\,|\Delta m| + 3.0\,|\Delta f| + 0.4\,N_{\text{float}} + 0.4\,N_{\text{width}} + \beta\, D_{\text{content}}(x')$$

$|\Delta m|$ = margin shift in mm; $|\Delta f|$ = font-size steps dropped; N_{float} = float-placement specs changed; N_{width} = figure-width expressions changed. Slackening is cheap by construction, so after hitting the target the system relaxes margins/fonts back to the "loosest optimal" point.

1.5 Feasibility gate and ordering

$$x' \text{ feasible} \iff L(x') = 0, \qquad J_1(x') = \begin{cases} +\infty, & L(x') > 0 \\ I(x'), & L(x') = 0 \end{cases}$$

Candidates are compared *lexicographically*: first L (0 beats any positive value), then I. The current implementation is the hand-written form `total = 1e6*len(lv) + a + i`; semantically equivalent, with 10^6 standing in for $+\infty$.

1.6 What changed vs. the previous "continuized L"

Term	previous (continuized)	revised (this version)
page term	$\max(0, N-b)/b$ (has a gradient)	0/1, over limit = infeasible
content term	continuous, still inside $\mathcal{H}$	moved out; soft βD_{content}
compile / specs / figures	0/1	0/1 (unchanged)

1.7 Trade-off to state explicitly: the objective has no gradient

With the page term at 0/1, L carries no search signal on the page axis. Therefore:

- **objective**: strictly 0/1 (non-negotiable, as required);
- **search heuristic**: during page compression use the residual $g(x') = \max(0, N_{\text{pages}}(x') - b)$ only to *choose/rank moves* and to advise the user ("0.3 pages to go"). g never enters L and never decides acceptance.

2. Model 2: aesthetic quantification A

2.1 Overall

$$\boxed{A(x') = A_{\text{defect}}(x') + \lambda\, A_{\text{profile}}(x')}, \qquad \lambda = 0 \text{ (shadow mode, current default)}$$

2.2 A_{defect}: detectable defects (the layer currently in force)

$$\begin{aligned} A_{\text{defect}} = {} & 1.0\, N_{\text{overfull}} + 0.6\, N_{\text{underfull}} + 1.0\, N_{\text{vboxOver}} + 1.4\, N_{\text{vboxUnder}} \\ & + 1.6\, N_{\text{floatwarn}} + 1.2\, N_{\text{tabOverwide}} + 0.3\, N_{\text{warn}} + 1.0\, N_{\text{figOverwide}} \\ & + 2.0\, \mathbf{1}[\text{quality macros missing}] \\ & + \sum_{fe} (0.8\, \mathbf{1}[\text{spec unstable}] + 0.5\, \mathbf{1}[\text{spec} = H]) + \sum_{h} w_{\text{hyg}}(h) + D_{\text{vis}} \\ & + 2.5\, \mathbf{1}[\text{TOC missing}] + 2.5\, \mathbf{1}[\text{running header missing}] + 1.5\, \mathbf{1}[\text{headings uncolored}] \end{aligned}$$

w_{hyg} = source-hygiene weights (`HYGIENE_WEIGHT`, set by fixability × impact); $D_{\text{vis}} = \sum_d \text{weight}(d) \cdot \text{severity}(d)$ over the six rendered-page defects (`giant_content`, `stranded_block`, `bottom_blank`, `mid_gap`, `near_empty`, `density_imbalance`).

2.3 A_{profile}: deviation from venue norms (shadow, $\lambda = 0$)

For page p with role $\text{role}(p) = r$, venue v, column count l, take the stratum $s = (v, r, l)$ profile

$$\{m_i,\ s_i,\ [\text{lo}_i, \text{hi}_i],\ R,\ \delta,\ \text{ref}_i,\ d2_{\text{ref}}\}$$

(median, IQR, correlation matrix):

1. heavy-tail transform $\tilde{x}_i = \log(1 + x_i)$ (positive-ratio dims only);
2. standardize $z_i = \text{clip}((\tilde{x}_i - m_i)/s_i, -4, +4)$;
3. two-sided out-of-band loss (below-lo and above-hi, asymmetric weights)

$$\text{loss}_i = W_{\text{low}} \frac{\max(0, \text{lo}_i - \tilde{x}_i)}{s_i} + W_{\text{high}} \frac{\max(0, \tilde{x}_i - \text{hi}_i)}{s_i}, \qquad W_{\text{low}} = 1.2 > W_{\text{high}} = 1.0;$$

4. per-dim normalization $\widehat{\text{loss}}_i = \min(\text{loss}_i/\text{ref}_i, 20)$, ref_i = own P90 of the stratum;
5. Mahalanobis $D^2 = z^\top \Sigma^{-1} z$ with $\Sigma = (1-\delta)R + \delta I$, δ by Ledoit–Wolf, and $\widehat{D^2} = D^2/d2_{\text{ref}}$;
6. page→document aggregation (CVaR-like)

$$\text{pool}(u) = 0.5 \cdot \text{topk}(u) + 0.5 \cdot \text{mean}(u), \qquad k = \max(1, \text{round}(0.05\, N_{\text{pages}}))\,;$$

7. composition

$$A_{\text{profile}} = \frac{\sum_i w_i \,\text{pool}(\widehat{\text{loss}_i}) + w_M \,\text{pool}(\widehat{D^2})}{\sum_i w_i + w_M}, \qquad w_i \equiv 1 \text{ (uncalibrated)},\ w_M = 1.$$

2.4 Honest boundaries

A_{profile} is a *ranking* quantity, not a human aesthetic score; it models only "layer 2: deviation from norms", not subjective preference; A_{defect} reads source + rendering while A_{profile} reads rendering only — they measure different things and can only be added, not substituted; λ stays 0 until its pre-gate calibration passes.

3. Where the formulas live in code (repo root: `/mnt/d//texopt/`)

Formula part	File	Function / constant
L violation list	`texopt/score.py`	`l_violations()` L309
L scale / composition	`texopt/score.py`	`L_PENALTY` L36, `total_score()` L446
Body normalization, Keep	`texopt/score.py`	`_norm_body()`, `body_unchanged()` L213, `semantic_diff()` L194
D_{content} soft term (new)	`texopt/score.py`	`content_deviation()` L409, `CONTENT_PENALTY` L39
Page-limit scope/judging	`texopt/score.py`	`page_status()`, `judge_pages()`, `within_limit()`
I intervention cost	`texopt/score.py`	`intervention_cost()` L425
A_{defect}	`texopt/score.py`	`aesthetic_score()` L368, `HYGIENE_WEIGHT` L42
D_{vis} (rendered pages)	`texopt/visual.py`	`visual_defects()` L416, `defect_penalty()` L423, `DEFECT_WEIGHT` L234, `analyze_pdf()` L386, `page_metrics()` L244
A_{profile} steps 1–5	`texopt/aesthetic.py`	`class Detector` L336 (vector L376, mahalanobis L407, band_loss L421, norm_loss L451, norm_d2 L457, score_page L464)
A_{profile} step 6	`texopt/aesthetic.py`	`pooled_agg()` L105, `topk_mean()`, `percentile()`
A_{profile} step 7	`texopt/aesthetic.py`	`evaluate_doc()` L633, `W_LOW/W_HIGH/W_D2` L43–45
Mahalanobis / shrinkage / χ^2	`texopt/aesthetic.py`	`corr_from_z()` L138, `lw_delta()` L164, `shrink()` L194, `chi2_sf()` L274, `chi2_isf()` L320
Profile construction	`texopt/aesthetic.py`	`build_block()` L481, `pick_detector()` L605
Weight calibration	`texopt/weights.py`	`a_profile_from_parts()` L136, `weights_from_trials()` L241, `ablation_weights()` L264, `fit_bradley_terry()` L348, `accept_verdict()` L318
Shadow-mode wiring	`texopt/shadow.py`	`evaluate_pdf()` L151, `snapshot()` L170, `gate_drop_dims()` L89

4. Change list for this revision

1. `Requirement.content_hard` (default `False`) — content-preservation policy switch.
2. `score.py`: new `CONTENT_PENALTY = 50.0` and `content_deviation()`; content check in `l_violations()` only active when `content_hard=True`; `total_score()` adds the soft term and returns `d_content` / `content_preserved`.
3. `tests/run_tests.py`: 4 new regression checks for the content policy.
4. Page limit: *no code change needed* — existing `if req.page_limit:` guard plus a 0/1 entry already matches directive 2.
5. Known minor bug (to fix): in `aesthetic.py::evaluate_doc()` the report fields `norm_d2_top` and `norm_d2_p90` hold the same value, as do `dim_norm_top` and `dim_norm_p90` — currently two indistinguishable columns.